

Name:- N. Punit Kumar

Reg NO:- 198505156

course code:- CSA0307

course Name:- Data structures

COS - AT1 - Pseudocode

1) Floyd-warshall with Reduced space
usage for GPs. path Reconstruction.

1) Floyd-warshall with path Reconst
PSEUDOCODE:- ion.

INPUT: Graph with N Locations.

output:- shortest distance b/w all
Locations.

create distance [N][N]

for i = 1 to N

for j = 1 to N

if i = j

distance [i][j] = 0

ELSE If edge exists from i to j

distance [i][j] = edge weight

ELSE

distance [i][j] = INFINITY.

next (i)(j) = -1

END IF

END FOR

END FOR

for k = 1 to N

for i = 1 to N


```

for j = 1 to N
  If distance(i)(k) + distance(k)(j) < (
    distance(i)(j) =
      distance(i)(k) + distance(k)(j)
    next [i)(j] = next [i)(k)
  END IF
END FOR
END FOR
END FOR

```

```

FUNCTION Getpath(source, destination)

```

```

  If next[source](destination) == -1

```

```

    RETURN "NO path"

```

```

  END IF

```

```

  path = empty list

```

```

  current = source

```

```

  ADD current TO path

```

```

  while current != destination.

```

```

    current = next[current](destination)

```

```

    ADD current TO path.

```

```

  END while

```

```

  RETURN path.

```

```

END function.

```


output distance

OUTPUT Get Path(source, destination)

END.

Ex:- shortest distance : 15 km

Actual shortest path:-

Location A $\rightarrow$ Location C $\rightarrow$ Location D
 $\rightarrow$ Location E.

space complexity :- $O(N^2)$ for distance
 $+ O(N^2)$ for next = $O(N^2)$ overall

Time complexity : $O(N^3)$.

(2) Floyd-warshall with Reduced space usage for GPS.

PSEUDOCODE :-

INPUT:- Graph with N Locations.

output:- shortest distance b/w all locations.

create distance[N][N]

for i = 1 to N.

for j = 1 to N

If $i == j$

distance[i][j] = edge_weight

else


```

distance[i][j] > INFINITY
END IF
END FOR
END FOR

FOR K = 1 TO N
  FOR I = 1 TO N
    FOR J = 1 TO N
      distance[i][j] = MIN(
        distance[i][j],
        distance[i][K] + distance[K][j]
      )
    END FOR
  END FOR
END FOR

output distance
END.

```

Space complexity : $O(N^2)$

memory saving: The same distance matrix is updated in-place